Epidemics on Contact Networks

Stochastic Transitions and Scenario Design

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1 Abstract

This MSc team project studies how heterogeneous sociability and contact structure affect a stochastic susceptible–exposed–infectious–removed model. Households, persistent non-household links and random contacts supply separate transmission routes. Mansi Neb, Riccardo Monforte and Fakhruddin Hakim Hussain share the recorded overall contribution.

The archived R script specifies four scenarios but has no preserved seeded numerical benchmark. The review therefore runs a separately labelled Python reference implementation: fifty replications per scenario, a population of 1,000, five initial infectious individuals and 100 daily states. It checks population conservation, state monotonicity, network symmetry and valid transmission probabilities in every run.

Mean peak infectious counts are 164.34, 208.44, 155.48 and 205.16 across the four cases. These are new conditional simulation results, not reproduced historical R outputs or forecasts for a real epidemic. Crucially, the random-only scenarios also increase random-route transmission from 0.01 to 0.04. The comparison does not isolate contact structure as a single causal factor.

The research decision is to retain the work as a stochastic modelling and scenario-design study. A cleaner experiment should match transmission assumptions, use a documented seed protocol and report variability over networks and epidemic paths before interpreting structural differences.

2 Population and Contact Model

The model assigns household membership by sampling household sizes from one to five and truncating the resulting membership vector to 1,000 individuals. Sociability β_i is either independently uniform on zero to one or constant at the population's sampled mean.

For different households, a persistent undirected edge is sampled with probability

$$p_{ij} = \min\left\{1, \frac{n_c \beta_i \beta_j}{\bar{\beta}^2 (n-1)}\right\}, \qquad n_c = 15. \tag{1}$$

Self-links and within-household links are excluded from this regular-contact graph. The upper-triangle draw is mirrored, ensuring symmetry. The value fifteen scales desired contact intensity; realised mean degree need not equal fifteen after exclusions and clipping.

Contacts and households are synthetic. There is no observed population network or calibration dataset in the supplied study. Realistic interpretation cannot be obtained by assigning a city name to these generated individuals.

3 Daily Transitions and Infection Probability

States are susceptible S, exposed E, infectious I and removed R. For each day, exposed individuals become infectious with probability 0.4 and previously infectious individuals become removed with probability 0.2. All updates use the previous day's states, so a new exposure cannot become infectious in the same update.

If a susceptible individual has k_h infectious household members and k_c infectious regular contacts, the route probabilities are

$$p_h = 1 - (1 - \alpha_h)^{k_h}, \qquad p_c = 1 - (1 - \alpha_c)^{k_c}. \tag{2}$$

For each infectious person i, the random-route probability is $m_{ji} = \alpha_r n_c \beta_j \beta_i / [\bar{\beta}^2 (n-1)]$. Conditional independence gives

$$p_r = 1 - \prod_{i \in I} (1 - m_{ji}), \qquad p_{\text{infection}} = 1 - (1 - p_h)(1 - p_c)(1 - p_r). \tag{3}$$

The implementation uses log-one-minus arithmetic for the random route and validates probabilities. These independence assumptions are part of the model, not observed properties of real social transmission.

4 Scenario Design and New Reference Runs

Table 1: Four archived scenario definitions.

Case	Sociability	Structured routes	Random coefficient
1	Heterogeneous	Included	0.01
2	Heterogeneous	Excluded	0.04
3	Constant	Included	0.01
4	Constant	Excluded	0.04

Within each replication, the household vector and sampled heterogeneous sociability define the four scenarios. Constant-sociability cases use that sample's mean. Networks and epidemic draws are stochastic. The master seed is 20261101, with replication indices zero through 49. The original R random-number stream is not reproduced.

The scenario comparison changes both contact structure and the random-route coefficient. This prevents attributing a difference solely to households or persistent contacts. Equal nominal coefficients would also not automatically match effective reproduction numbers; a future controlled experiment must state what transmission quantity is held constant.

5 Monte Carlo Results and Checks

Table 2: New Python reference simulations, fifty paths per case.

Case	Mean peak I	95% interval for mean	Mean ever infected
1	164.34	[160.15, 168.53]	838.40
2	208.44	[203.96, 212.92]	823.36
3	155.48	[151.42, 159.54]	879.66
4	205.16	[201.03, 209.29]	938.52

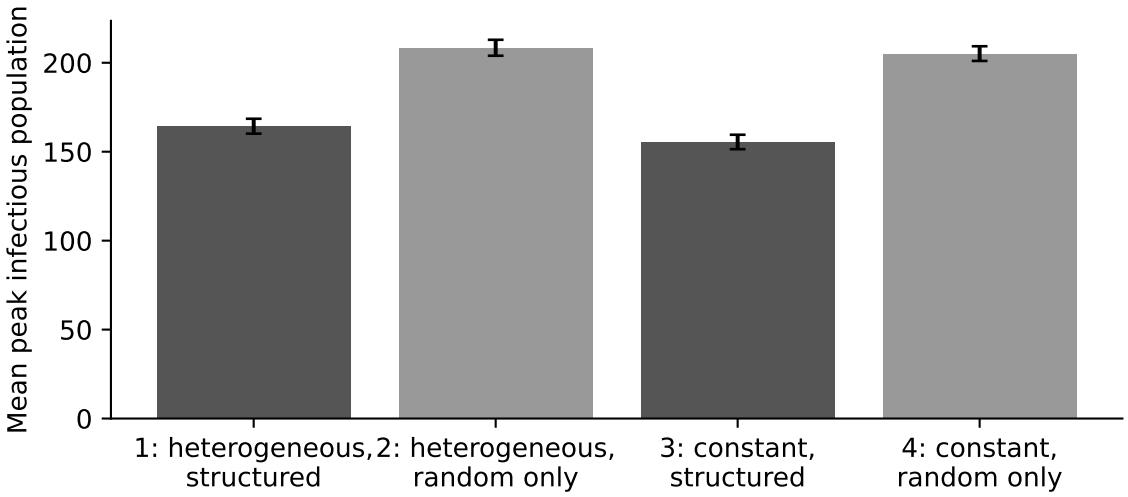

Figure 1: New Python reference results. Bars are means over fifty paths; error bars describe approximate 95% Monte Carlo intervals for the mean, conditional on these fixed scenario rules.

Every one of the 200 epidemic paths conserves the population at every time step. Susceptibles never increase, removed counts never decrease, contact networks are symmetric with no self/household edges, and all computed transmission probabilities are valid. These checks establish basic execution invariants, not model calibration or external predictive performance.

Ever infected means $n - S_{99}$, including the initial five infectious individuals and any exposed or infectious people remaining at the horizon. It is not identical to the final removed count when transmission has not entirely ended. Peak timing and peak height are random summary statistics and should not be promoted from one visually selected run.

6 Conclusion and Reproducibility

The project demonstrates stochastic state updates, network construction and the need to separate a model mechanism from an experimental contrast. The new reference runs give reproducible examples and make the scenario confounding explicit. They are not a reproduction of missing original figures or a public-health recommendation.

The next experiment should vary one structural factor at a time, predeclare an intensity-matching rule and separate variability from household formation, networks and epidemic paths. Sensitivity to population size, initial conditions, transmission strengths and run length is required before broader conclusions. All new outputs remain conditional teaching simulations.

The original R analysis remains available. The public Python reference runner and aggregate JSON provide the new seed, settings and summaries. An R runtime was unavailable in this review; no R execution is claimed.

References

[1] Project authors. Pinned source code and evidence. Original study distinguished from the November 2026 audit.

[2] Companion verification record. Seed, inputs, numerical scope and limitations.